approved per billion US dollars halves over a fixed period of time, and is visualized in Figure 1.3 from a Nature paper (https://www.nature.com/articles/nrd3681), where drug discovery throughput resulting from each billion-dollar investment decreases on the logarithmic scale.

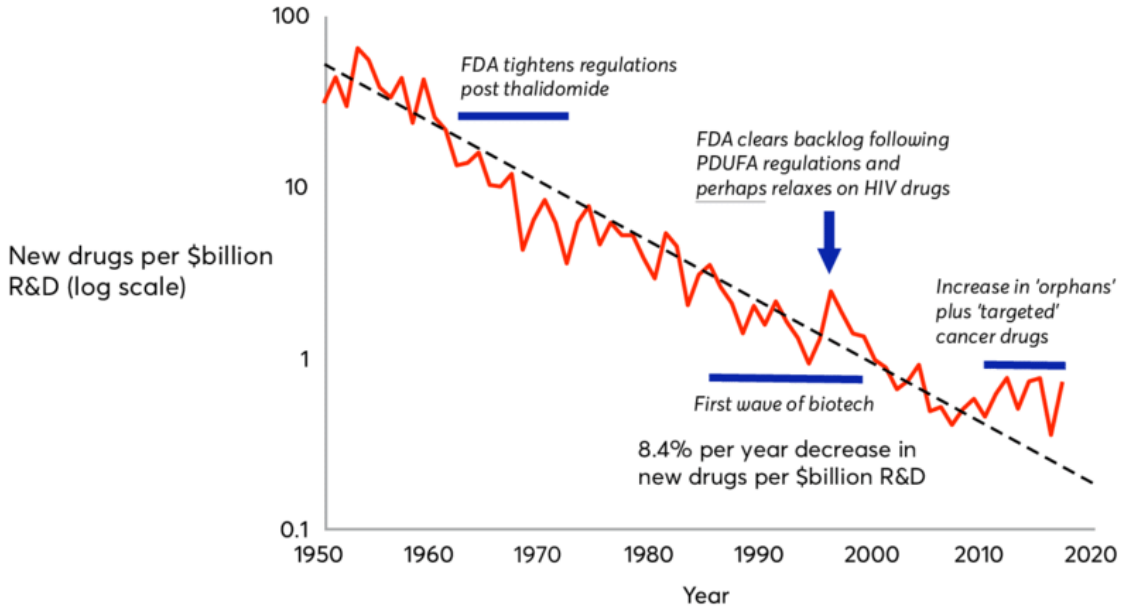


Figure 1.3 Eroom's law dictates that it will cost increasingly more money to produce effective drugs. Therefore, drug discovery is also a fundamentally hard problem.

The same problem in fact applies to any scientific discovery task, where scientists search for new chemicals, materials, or designs that are rare, novel, and useful with respect to some metric, using experiments that require top-of-the-line equipment and may take days or weeks to finish. In other words, they are trying to optimize for their respective objective functions where evaluations are extremely expensive.

Table 1.2 Data from a materials discovery task, extracted from the author's research work. The objective is to find the alloy composition (from the four parent elements) with the lowest mixing temperature, which is also a black-box optimization problem.

% of Pb	% of Sn	% of Ge	% of Mn	Mixing temp.
0.50	0.50	0.00	0.00	192.08
0.33	0.33	0.33	0.00	258.30
0.00	0.50	0.50	0.00	187.24
0.00	0.33	0.33	0.33	188.54

As an illustration, Table 1.2 shows a couple of data points in a real-life data set from such a task. Here, materials scientists worked with alloys that are of lead (Pb), tin (Sn), germanium (Ge), and manganese (Mn) compositions. Each given combination of percentages of these compositions corresponds to a potential alloy that could be synthesized and experimented on in a laboratory.

©Manning Publications Co. To comment go to liveBook
https://livebook.manning.com/#!/book/bayesian-optimization-in-action/discussion

Licensed to ashwin sampath <ashwin.manning@gmail.com>